

Air Pollution’s Grip: Drug Expenditure and Its Heterogeneity in China*

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Abstract

We examine how air pollution translates into pharmaceutical expenditure in China, accounting for both Western medicine (WM) and Chinese herbal medicine (CHM). Using drug-level expenditure and quantity data from public hospitals in China during 2014–2019, we study medicines used to treat respiratory diseases, cardiovascular diseases, and tumors. We estimate pollution elasticities of product-level per-capita expenditure ranging from 2.96 to 4.91 for WM and from 4.35 to 5.57 for CHM across the three disease groups. A 1% increase in annual average PM_{2.5} increases combined expenditures on drugs for the three disease groups by an amount equivalent to 1.81% of annual per-capita drug expenditure. The estimated CHM expenditure responses are comparable in magnitude to those for WM, highlighting a component of pollution-related pharmaceutical spending that has received limited attention. Higher pollution is also associated with increased drug quantities across all six disease-medicine groups. These findings suggest that greater medication use accompanies the expenditure response and underscore the importance of including CHM when assessing the pharmaceutical burden of air pollution in China.

JEL classification: I15, Q53

Keywords: outdoor air pollution, drug expenditure, Chinese herbal medicine, disparities in drug expenditure

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1 Introduction

Outdoor air pollution imposes substantial health burdens, particularly through respiratory illnesses, cardiovascular diseases, and tumors.¹ Existing research has documented several categories of economic costs associated with exposure to pollution—such as defensive expenditures (Deschenes, Greenstone, and Shapiro, 2017; Zhang and Mu, 2018; Ito and Zhang, 2020), drug expenditures (Williams et al., 2019), and hospitalization costs (Schlenker and Walker, 2016; Deryugina et al., 2019). However, most of this evidence comes from developed countries (Currie and Neidell, 2005; Schlenker and Walker, 2016; Jans, Johansson, and Nilsson, 2018; Deryugina et al., 2019; Giaccherini, Kopinska, and Palma, 2021; Margaryan, 2021; Alexander and Schwandt, 2022; von Hinke and Sørensen, 2023; Bishop, Ketcham, and Kuminoff, 2023), with much less attention paid to middle- and lower-income settings, despite these regions experiencing more severe pollution and weaker health protection.

Although prior studies have examined the mortality consequences of pollution in China (Ebenstein et al., 2015; He, Fan, and Zhou, 2016; He, Liu, and Zhou, 2020) and India (Greenstone and Hanna, 2014), the broader economic costs of exposure to pollution in China remain insufficiently understood. A further gap concerns the role of Chinese herbal medicine (CHM), which is widely used in China to treat pollution-related illnesses, but is largely absent from the health-economics literature.² Consequently, both the economic burden of air pollution in China and the usage patterns of CHM remain understudied. Our study directly addresses these gaps by quantifying the response of drug-expenditures to pollution and providing the first systematic evidence on the contribution of CHM expenditures to the economic burden of air pollution.

The existing literature identifies several interconnected pathways through which air pollution translates into drug expenditures. At the physiological level, exposure to fine particulate matter (PM_{2.5}) triggers systemic inflammation, oxidative stress, and impaired respiratory and cardiovascular function, increasing the incidence and severity of pollution-related diseases and

¹According to the World Health Organization, outdoor air pollution was estimated to cause 4.2 million premature deaths worldwide in 2019 (see [https://www.who.int/zh/news-room/fact-sheets/detail/ambient-\(outdoor\)-air-quality-and-health](https://www.who.int/zh/news-room/fact-sheets/detail/ambient-(outdoor)-air-quality-and-health)). Coarse particles with a diameter greater than 2.5 microns can penetrate the lungs, while fine particles can cross the lung barrier and enter the bloodstream (Chang et al., 2019). Exposure to fine particulate matter of 2.5 microns or less in diameter (PM_{2.5}) can lead to respiratory diseases, cardiovascular diseases, and tumors (Seaton et al., 1995).

²There exists a nontrivial preference for CHM drugs, as on average 13.5% of hospital visits occur at Traditional Chinese Medicine hospitals in the provinces to which these cities belong.

thus increasing the clinical demand for medication (Seaton et al. 1995; Miller et al. 2020; Miller, Wherry, and Mazumder 2021; Aryal, Harmon, and Dugas 2021). At the behavioral level, households respond to perceived health risks by making “defensive investments” in health, including greater use of symptom-relief and maintenance drugs, stockpiling of medications around high-pollution episodes, and shifting toward preparations believed to strengthen immunity. Such behavioral patterns have been documented in both developed and developing country settings (Deschenes, Greenstone, and Shapiro, 2017; Barwick et al., 2018). Institutionally, air pollution increases outpatient visits and hospital admissions, and the structure of public insurance and hospital financing determines how this additional demand is converted into observed pharmaceutical spending (Schlenker and Walker, 2016; Deryugina et al., 2019; Liao, Du, and Chen, 2021).

Using a novel dataset that records drug-level expenditures from public hospitals in 20 major Chinese cities from 2014 to 2019, we estimate how established pollution-related health effects translate into pharmaceutical expenditures. Specifically, we examine how expenditures on Western Medicine (WM) and Chinese Herbal Medicine (CHM) respond to changes in air quality, and how socioeconomic factors shape the sensitivity of cities’ drug expenditures to pollution-related health risks.

To address potential endogeneity, we estimate the relationship using an instrumental-variable approach. First, air pollution imposes a substantial economic cost in terms of drug expenditures. A 1% increase in the annual average of PM2.5 results in a 2.96% increase in per capita expenditure on modern respiratory drugs, a 3.59% increase in expenditure on modern cardiovascular drugs, and a 4.91% increase in per capita expenditure on modern antitumor drugs. The corresponding combined increase in drug expenditures on WM drugs amounts to 9.10 yuan, equivalent to 1.00% of the annual per capita drug expenditure in our sample.

Second, CHM expenditures constitute a quantitatively important part of the pharmaceutical expenditure response to air pollution.³ Following a 1% increase in PM2.5, per capita expenditures on CHM drugs for treating the three groups of diseases increase by 5.57%, 4.35% and 4.76%, respectively. In terms of increases in expenditures, the corresponding increase in expenditures on CHM drugs (for respiratory diseases, cardiovascular diseases, and tumors combined)

³The WM drugs in our sample are classified into 14 groups, while the CHM drugs are classified into 13 groups; The classifications are detailed in Table 1.

is 7.44 yuan, which is equivalent to 0.81% of the annual expenditure per capita of drugs.

Third, the socioeconomic characteristics of cities shape how air pollution translates into drug expenditures, contributing to regional disparities in pollution-related healthcare costs. Our heterogeneity analysis suggests that this relationship depends on differences in healthcare access and the ability to finance medical treatment across cities. Specifically, the estimated response of WM drug expenditures is larger in cities with a higher share of college graduates, with suggestive evidence of a larger response in cities with higher per capita disposable income.

In policy terms, these results indicate that improving air quality is a potentially high-return public investment with measurable health system gains. Besides improving health conditions of the population, better air quality can reduce public and private spending on drugs, and the pollution-amplified disparity in healthcare usage.

Our study contributes to the related literature in three ways. First, while most articles estimate the effects of air pollution on drug expenditures by disease groups, such as respiratory diseases, cardiovascular diseases (Schlenker and Walker, 2016; Xia et al., 2022), psychological disorders (Chen, Oliva, and Zhang, 2018), and flu (Graff Zivin et al., 2023), our study offers a more comprehensive assessment of the impact of air pollution on drug expenditures in three disease groups considered the most susceptible to air pollution (Seaton et al., 1995). In the context of China, our study is related to research by Barwick et al. (2018) and Liao, Du, and Chen (2021), who estimate annual per capita medical spending using data on credit and debit card transactions and the China Family Panel Studies dataset, respectively. Drawing on a new data source, our study provides a specific focus on drug expenditures. By exploiting variation in air pollution and drug expenditures across major Chinese cities, we also improve upon Zhang et al. (2023) and Xia et al. (2022), who base their estimates of drug expenditure responses to air pollution on data from a single city.

Second, we highlight the importance of CHM drugs in the medical expenditures and practices of China. Despite accounting for a substantial share of spending on treatments for these three disease groups in the sampled pharmaceutical markets, CHM drugs remain relatively understudied in terms of their efficacy and pharmacoeconomic evaluation (Zhou et al., 2019; Xiong et al., 2022). Examining the expenditure responses to air pollution, we demonstrate that the increase in expenditure on CHM drugs is comparable in magnitude to that on WM drugs.

To the best of our knowledge, this study is the first to examine the usage patterns of CHM drugs. This finding has important implications for expanding research on CHM drugs.

Third, we contribute to the research on regional disparities in the impact of air pollution. Existing literature has demonstrated that locations with better socioeconomic status are less affected by air pollution. In the context of the U.S., a key finding by Currie, Voorheis, and Walker (2023) is that exposure to air pollution tends to be lower in communities with higher rates of home ownership, greater average educational attainment, and higher mean public assistance income. Exploiting variation in early childhood exposure to the London Smog of 1952, von Hinke and Sørensen (2023) find that individuals born in areas with higher socioeconomic status suffer fewer long-term effects from air pollution, presumably because they are better at avoiding pollution or possess superior health conditions. Our findings contribute to understanding how socioeconomic factors moderate the impact of pollution by revealing that Chinese cities with higher socioeconomic status spend more on drugs in response to air pollution.

A related geographic dimension concerns the well-documented north–south disparity in the health effects of air pollution in China. The literature on air pollution in China (Chen et al., 2013; Ebenstein et al., 2017; Fan, He, and Zhou, 2020) has established that residents in northern China suffer from lower life expectancy because the free or subsidized coal-fired heating provided in northern cities during the winter causes severe air pollution. We complement this strand of literature by showing that there is also a higher economic cost in northern cities, as residents there spend more to treat pollution-related diseases.

The remainder of the paper is structured as follows. Section 2 presents the empirical strategy and data. Section 3 reports the benchmark results. Section 4 examines heterogeneity in the expenditure response across cities with different socioeconomic characteristics. Section 5 presents robustness checks and extensions. Section 6 concludes.

2 Empirical Design

2.1 Regression setup

We employ the following log-log specification to estimate how variations in air pollution translate into changes in drug expenditures:

$$\ln Y_{ijt} = \alpha + \beta \cdot \ln PM2.5_{it} + \phi \cdot \ln X_{it} + \gamma_i + \eta_t + \epsilon_{ijt} \quad (1)$$

where Y_{ijt} is per capita expenditure on drug j in city i during year t . In our study, we focus on three groups of diseases, the respiratory diseases, cardiovascular diseases and tumors.

By the origin of medicine practice, there are two types of drugs. One type is the modern chemical drugs used in WM, such as penicillin, which are sold in countries around the world. The other type is CHM drugs, mainly sold and used in China and Japan. CHM drugs primarily originate from natural substances and their processed products, including herbs, animal-based drugs, mineral drugs, as well as certain chemical and biological products. The use of CHM drugs is widespread in China. Among the 685 drugs listed in China’s National Essential Medicines List (2018 edition), 268 are CHM drugs, while 417 are WM drugs. For instance, “Huangqi Granules”, produced by “Sichuan Biokin Pharmaceutical”, are derived from the roots of *Astragalus membranaceus* (Fisch.) and Bunge. This CHM drug is commonly used to enhance immune function. National and provincial-level health authorities regularly issue guidelines regarding the inclusion of CHM drugs in public health insurance programs in China. The classifications of WM drugs and CHM drugs in our sample, reported in Table 1, largely overlap.

Table 1: Grouping of WM drugs and CHM drugs

WM drugs	CHM drugs
Identical grouping:	
Antitumor drugs	Antitumor drugs
Cardiovascular drugs	Cardiovascular drugs
Respiratory drugs	Respiratory drugs
Dermatological drugs	Dermatological drugs
Neurological drugs	Neurological drugs
Musculoskeletal drugs	Musculoskeletal drugs
Hematological drugs	Hematological drugs
Gastrointestinal drugs	Gastrointestinal drugs
Miscellaneous drugs	Miscellaneous drugs
Differentiated grouping:	
Anti-infective drugs	Otorhinolaryngology drugs
Systemic Hormonal Preparations	Pediatric drugs
Sensory System drugs	Gynecological drugs
Antiparasitic, Insecticide, and Repellent drugs	Urological Disorder drugs
Genitourinary and Sex Hormones	

Notes : [1] The classification of Western medicine drugs follows the Anatomical Therapeutic Chemical (ATC) classification developed by the World Health Organization. The classification of CHM drugs is developed by the MENET Database, with reference to the ATC classification. [2] Systemic Hormonal Preparations in WM drugs excluding Sex Hormones and Insulins.

CHM drugs play a crucial role in our analysis, as the per capita expenditure on them is approximately 75% of the expenditure on WM drugs for treating respiratory diseases, cardiovascular diseases, and tumors. Since our sample includes both WM and CHM drugs for treating

respiratory diseases, cardiovascular diseases and tumors, we conduct separate regressions for each of the six combinations of origin of medicine and disease group.

The variable $PM2.5_{it}$ represents the mean PM2.5 concentration in city i during regression year t . To align the measurement periods for pollution exposure and drug expenditures, we define regression year t as the period from the fourth quarter of calendar year $t - 1$ through the third quarter of calendar year t . This definition is applied consistently to both drug expenditures and pollution exposure. For example, regression year 2018 covers drug expenditures and pollution exposure from the fourth quarter of 2017 through the third quarter of 2018.

X_{it} is a vector of control variables. Within X_{it} , the first set of controls comprises climate-related indicators, including the within-period average precipitation, average humidity, and average temperature. The second set of controls consists of socioeconomic characteristics of the cities: within-period per capita GDP in city i , fiscal revenue in city i , the fraction of urban residents covered by employment-based health insurance in city i , and the number of public hospitals at the provincial level. The benchmark specification includes city and year fixed effects. Our main inference procedure clusters standard errors at the city level.

2.2 Data

2.2.1 Drug Expenditures

We obtain quarterly expenditure and quantity consumed data from public hospitals for WM drugs across 20 major Chinese cities and CHM drugs across 9 major Chinese cities between 2014 and 2019 from the MENET database.⁴ Our main dependent variable is annual per capita expenditure on a drug in a city’s public hospitals. Each drug is identified by its commercial product name, dosage, package, and producer. For example, an observation in the original data represents the total expenditure in Beijing in the fourth quarter of 2019 on Felodipine Tablets” (dosage: 5 mg; package: 28 tablets) produced by Beijing Union Pharmaceutical Factory.” We aggregate the quarterly observations to the annual frequency.

The 20 cities are Beijing, Changchun, Changsha, Chengdu, Chongqing, Fuzhou, Guangzhou, Hangzhou, Harbin, Jinan, Nanjing, Shanghai, Shenyang, Shenzhen, Shijiazhuang, Taiyuan, Tianjin, Wuhan, Xi’an, and Zhengzhou. Their combined population was 253.82 million at the end of 2019. Quarterly expenditure and quantity data for CHM drugs are available for

⁴Source: <https://www.menet.com.cn/menetDatabase/dbCountyHospital.html>.

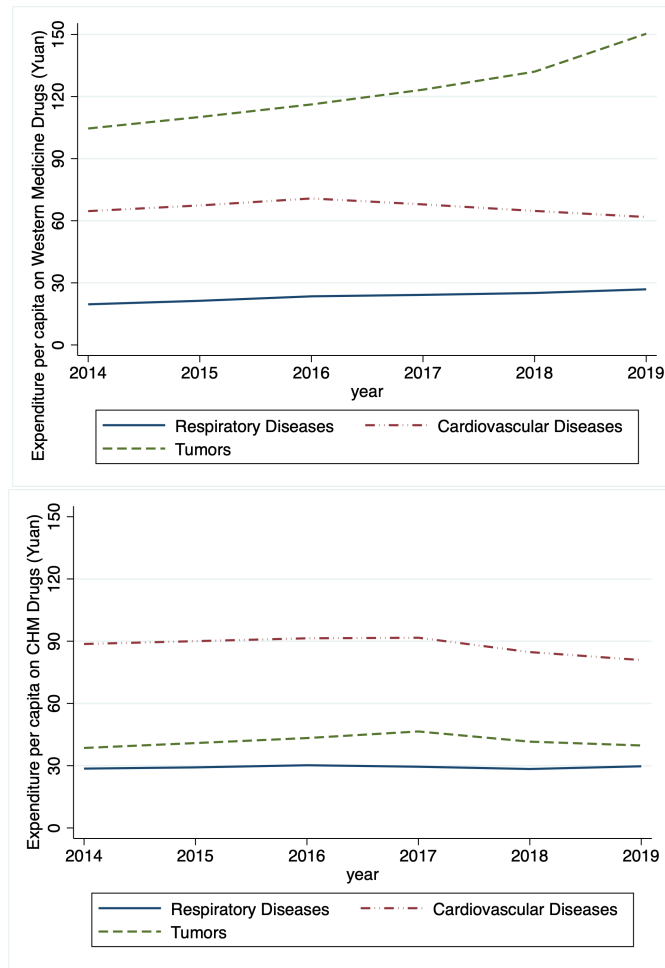
nine cities (Beijing, Chengdu, Chongqing, Guangzhou, Harbin, Nanjing, Shenyang, Xi'an, and Zhengzhou) over the same period, with a combined population of 130.79 million. The WM and CHM datasets are collected from 1,805 public hospitals, accounting for 60.17% of all public hospitals in China. As shown in Section J of the Online Appendix, the sample cities are generally larger and more developed than other Chinese cities, although their advantages in healthcare access are not consistent across all measures. Accordingly, we interpret our estimates as evidence for major urban pharmaceutical markets.

To mitigate the impact of outliers, observations with annual per capita expenditure and annual average prices falling below the 1st percentile or exceeding the 99th percentile are excluded. Consequently, we retain 28,636, 49,110, and 47,205 observations for Western Medicine (WM) drugs associated with respiratory diseases, cardiovascular diseases, and tumors, respectively. Similarly, for CHM drugs in the same categories, we retain 8,748, 13,249, and 4,033 observations.

In Figure 1, the time series of nominal expenditures on WM drugs for the treatment of respiratory diseases and cardiovascular diseases exhibit no discernible trends, while expenditure on antitumor drugs demonstrates an upward trajectory. In contrast to WM drugs, the per capita expenditure on cardiovascular diseases is higher than that on tumors for CHM drugs. The expenditures remain stable over the sample period, signifying the enduring significance of CHM drugs in medical practice in China. Locations of the cities, together with drug expenditures, are shown in Figure 2.⁵

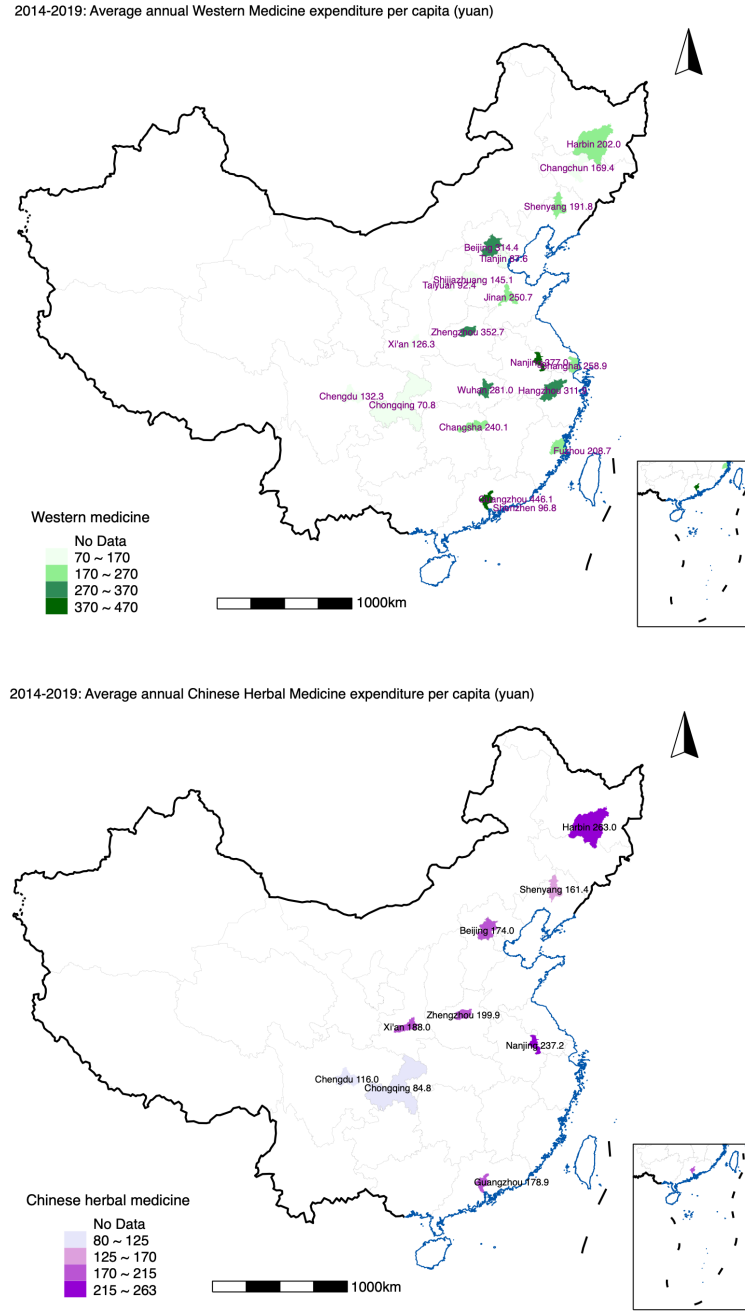
⁵Figure OA1 of the Online Appendix illustrates the per capita drug expenditure for each type of drug in all cities in 2019. There is a pronounced regional disparity in per capita expenditure for both WM drugs and CHM drugs. Notably, in 2019, the city with the highest GDP per capita (Nanjing, 165,294 yuan) exceeded the expenditure of the city with the lowest GDP per capita (Shenyang, 40,324 yuan) by 310%.

Figure 1: Expenditures on Western Medicine and Chinese Herbal Medicine Drugs



Notes : The top panel and bottom panel plot the current price expenditures per capita on WM drugs and CHM drugs, respectively. Source: authors' calculation.

Figure 2: Maps of City-level Expenditures on WM drugs and CHM drugs



Notes : The top (bottom, respectively) panel shows locations of cities and combined current price expenditures per capita on Western medicine drugs (Chinese herbal medicine) for treating respiratory diseases, cardiovascular diseases and tumors, averaged between 2014 to 2019. Source: authors' calculation.

2.2.2 Air Pollution and Control Variables

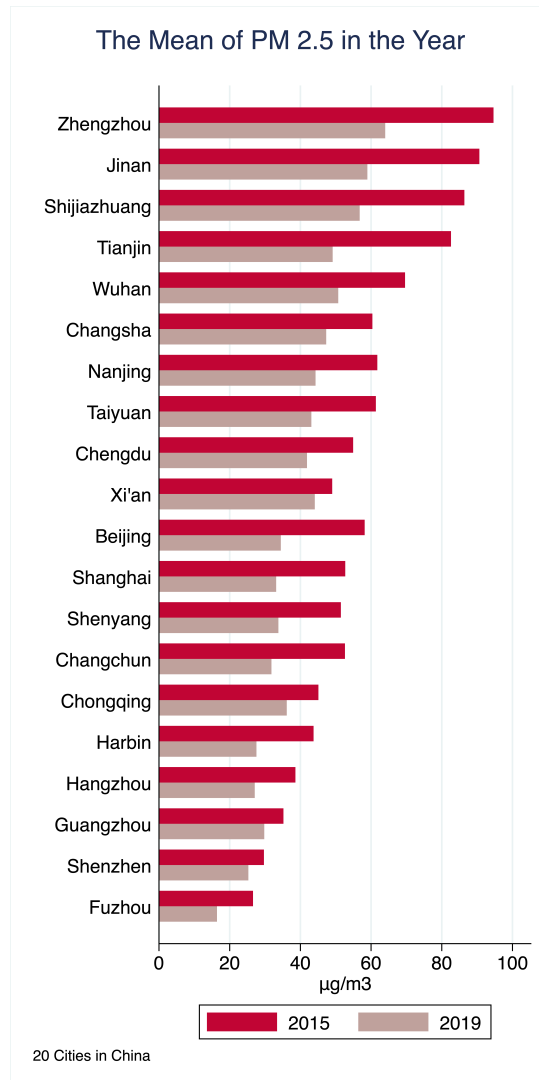
Our measure of air pollution, PM2.5, is obtained from the Tracking Air Pollution program, which integrates satellite-based observations from the National Aeronautics and Space Administration (NASA) with data from ground-based monitoring facilities in China.⁶ In the robustness checks, we also use the PM2.5 data from the Air Quality Index (AQI) published by the Chinese government, derived from readings obtained through a network of ground-based detection facilities. Figure 3 illustrates the substantial variation in air pollution levels across cities. Simultaneously, air quality has demonstrated improvement in all cities over time. From the map in Figure 4, we can see that the pollution level is higher in northern China on average.

To account for weather conditions, we have retrieved the daily values of temperature, precipitation, and humidity from representative meteorological stations in each city through the National Weather Bureau. We include their annual averages in the regressions.

As for socioeconomic characteristics, we incorporate the following indicators: per capita GDP from the National Bureau of Statistics, fiscal revenue from WIND datasets, the fraction of urban residents covered by employment-based health insurance from the China City Statistical Yearbook, and the number of public hospitals at the provincial level from the China Health Statistical Yearbook. A summary of the key variables is provided in Table 2. In addition to the previously noted disparities in drug expenditure, there exists substantial variation in key socioeconomic indicators such as GDP per capita, fiscal revenue per capita, household income, and educational attainment.

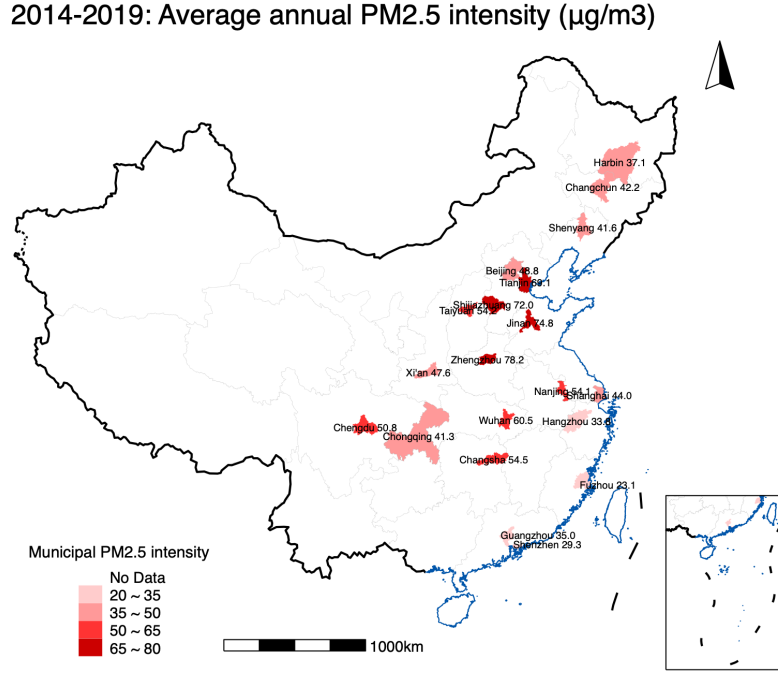
⁶The Tracking Air Pollution in China initiative is managed by a group of researchers at Tsinghua University, and the data is accessible at <http://tapdata.org.cn>.

Figure 3: City-level PM2.5



Notes : This figure illustrates the annual average PM2.5 of 20 cities in 2015 and 2019. The PM2.5 data are based on readings from a system of ground-based facilities and the information from NASA satellites. Source: authors' calculation.

Figure 4: Map of City-level PM2.5



Notes : The map shows locations of cities and the level of PM2.5 averaged between 2014 to 2019. Source: authors' calculation.

3 Benchmark Results

3.1 OLS estimates of effects of air pollution on drug expenditure

The OLS estimates in Table 3 show a consistent positive relationship between annual PM2.5 concentrations and drug expenditures across all three disease groups. For respiratory diseases, a 1% increase in annual average PM2.5 is associated with increases of 0.95% and 1.62% in per capita expenditures on WM and CHM drugs, respectively (Columns (1) and (3)). The corresponding estimates are 0.83% and 2.00% for cardiovascular drugs (Columns (5) and (7)), and 1.27% and 1.93% for antitumor drugs (Columns (9) and (11)). Notably, antitumor drugs constitute the largest expenditure category among WM drugs and the second largest among CHM drugs (Table 1). Across all three disease groups, the estimated association is larger for CHM than for WM drugs.

Table 2: Summary Statistics

Variable	N	Mean	SD	Min	Max
Dependent Variables :					
Annual expenditure per capita					
WM drugs: respiratory diseases (yuan)	28,636	0.067	0.157	0	1.338
CHM drugs: respiratory diseases (yuan)	8,748	0.131	0.372	0	3.911
WM drugs: cardiovascular diseases (yuan)	49,110	0.113	0.308	0	2.681
CHM drugs: cardiovascular diseases (yuan)	13,249	0.317	0.695	0	5.482
WM drugs: tumors (yuan)	47,205	0.246	0.522	0	3.794
CHM drugs: tumors (yuan)	4,033	0.526	1.026	0	7.008
Measures of Air Pollution					
Annual PM2.5 in TAP data, mean ($\mu g/m^3$)	140	52.290	19.104	16.410	106.205
Annual PM2.5 in TAP data, maximum ($\mu g/m^3$)	140	182.843	63.584	47.967	334.362
Annual PM2.5 in AQI data, mean ($\mu g/m^3$)	120	54.238	17.138	23.964	100.215
Annual PM2.5 in AQI data, maximum ($\mu g/m^3$)	120	263.486	141.669	72.375	903.792
Weather Controls					
Annual precipitation (mm)	140	3.188	1.043	1.366	5.970
Annual average relative humidity (%)	140	66.385	9.905	47.265	83.158
Annual average temperature (Celsius)	140	15.661	4.621	4.302	24.046
Socioeconomic Factors					
Annual GDP per capita (yuan)	140	88335.570	33034.430	13744	165294
Number of public hospitals in the province	133	1151.797	547.062	328	2615
Percentage of employee subscribed to basic health insurance over residents(%)	140	0.465	0.538	0.004	4.525
Annual fiscal revenue per capita (yuan)	140	11436	6438	3001	29509
Household income (yuan)	133	40823	11310	23058	73849
Fraction of residents with college education (%)	140	16.001	8.473	7.730	50.490

Notes : [1] Each drug is identified by commercial product name, dosage, package, and producer. For example, an observation is the total expenditure on ‘Felodipine Tablets’ (size:5mg; packaging: 28 tablets) produced by ‘Beijing Union Pharmaceutical Factory’ in the city of Beijing in 2019. [2] The source of Air Quality Index (AQI) data is prefecture-level governments in China. The source of TAP data is the Tracking Air Pollution in China program run by a group of researchers at Tsinghua University.

3.2 Two-Stage Least Squares estimates

OLS estimates may be biased by omitted variables, as local economic conditions, healthcare resources, or unobserved disease prevalence may be correlated with both air pollution and drug expenditures. To address this concern, we estimate the model using an instrumental-variable (IV) approach based on exogenous meteorological variation in air pollution.

Specifically, we use temperature inversion and maximum wind speed as instruments for PM_{2.5}. Temperature inversion refers to a meteorological condition in which air temperature near the ground is lower than that in the atmospheric layer above. These instruments satisfy the relevance condition through well-established atmospheric mechanisms: stronger temperature inversions suppress vertical air mixing and increase pollution concentrations, whereas stronger winds disperse pollutants. The exclusion restriction requires that, conditional on annual weather conditions and city and year fixed effects, these meteorological variables affect drug expenditures primarily through their effects on pollution exposure. The instruments have strong predictive power for PM_{2.5}, with first-stage F-statistics exceeding 300.

To examine sensitivity to the pollution measure and instrument choice, Table 8 reports estimates using alternative PM_{2.5} measures, and Online Appendix Section G reports estimates using temperature inversion as the sole instrument.

The 2SLS estimates in Table 3 are positive and statistically significant across all three disease groups. A 1% increase in annual average PM_{2.5} is associated with increases of 2.96% and 5.57% in per capita expenditures on WM and CHM respiratory drugs, respectively (Columns (2) and (4)). The corresponding estimates are 3.59% and 4.35% for cardiovascular drugs (Columns (6) and (8)), and 4.91% and 4.76% for antitumor drugs (Columns (10) and (12)).

Table 3: Effects of Air Pollution on Drugs Consumption

	Respiratory Tract Diseases				Cardiocerebral Vascular Diseases				Tumors			
	WM		CHM		WM		CHM		WM		CHM	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	OLS	2SLS	OLS	2SLS	OLS	2SLS	OLS	2SLS	OLS	2SLS	OLS	2SLS
ln(PM2.5)	0.952*** (0.251)	2.963** (1.142)	1.623*** (0.267)	5.571** (2.072)	0.829*** (0.060)	3.592** (1.628)	1.998*** (0.167)	4.347* (2.039)	1.268*** (0.375)	4.910* (2.402)	1.928*** (0.364)	4.763* (2.230)
ln(Precipitation)	0.196** (0.078)	0.522** (0.198)	0.415*** (0.128)	1.319** (0.453)	0.204*** (0.039)	0.587** (0.243)	0.416*** (0.031)	1.098* (0.569)	0.274** (0.119)	0.701** (0.326)	0.567** (0.227)	1.082* (0.475)
ln(Humidity)	-0.181 (0.233)	0.288 (0.552)	-0.274 (0.393)	-0.700 (1.017)	-0.145* (0.083)	-0.066 (0.570)	-0.798*** (0.023)	-0.586 (1.057)	-0.217 (0.327)	0.033 (0.697)	-0.712* (0.365)	-1.262 (0.994)
ln(Temperature)	0.516*** (0.200)	2.536** (1.096)	0.894*** (0.225)	4.176** (1.765)	0.573*** (0.027)	2.905* (1.402)	1.284*** (0.077)	3.256* (1.516)	0.863*** (0.272)	4.050* (2.063)	1.072*** (0.326)	3.516* (1.680)
ln(GDP per capita)	2.638*** (0.326)	5.035*** (0.420)	2.713*** (0.416)	4.972*** (0.641)	2.696*** (0.167)	5.329*** (0.523)	3.260*** (0.198)	6.414*** (0.856)	3.506*** (0.504)	6.047*** (0.737)	3.707*** (0.742)	6.929*** (0.914)
ln(Fiscal revenue)	0.282 (1.763)	1.252 (4.160)	0.830 (2.259)	2.081 (6.094)	-0.316 (0.406)	1.493 (4.717)	-1.886*** (0.461)	-1.355 (5.818)	1.473 (2.195)	3.542 (5.845)	2.274 (3.393)	1.269 (6.580)
ln(Insurance)	0.216*** (0.059)	0.611*** (0.157)	0.320*** (0.072)	0.946*** (0.254)	0.248*** (0.048)	0.668*** (0.198)	0.490*** (0.086)	1.010*** (0.232)	0.379*** (0.067)	0.826*** (0.225)	0.669*** (0.068)	1.210*** (0.251)
ln(Hospital)	1.161** (0.477)	1.518 (1.114)	0.121 (0.487)	-0.921 (0.876)	1.138*** (0.090)	1.501 (1.174)	0.331*** (0.119)	0.455 (0.945)	1.572*** (0.557)	2.051 (1.402)	1.392* (0.760)	1.076 (1.054)
Constant	-12.243*** (4.793)	-8.514** (3.613)	-8.514** (3.613)	-8.514** (3.613)	-11.624*** (0.434)	-11.624*** (0.434)	-10.196*** (1.638)	-10.196*** (1.638)	-16.849*** (5.337)	-16.849*** (5.337)	-16.769*** (5.290)	-16.769*** (5.290)
City FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Var. Mean	2.423	2.423	2.957	2.957	2.617	2.617	3.952	3.952	3.623	3.623	4.667	4.667
CD F statistic (first-stage)		1271.497		559.219		1995.921		942.197		1842.739		277.857
KP F statistic (first-stage)		6.431		5.205		6.178		6.643		5.912		6.437
Within R ²	0.224		0.268		0.198		0.267		0.247		0.265	
N	28,636	28,636	8,748	8,748	49,110	49,110	13,249	13,249	47,205	47,205	4,033	4,033

Notes : [1] Table reports OLS and 2SLS estimates of equation (1) from the main text. [2] Dependent variable is the log of per capita expenditure on drug j for treating respiratory diseases, cardiocerebral vascular diseases and tumors in city i in year t . The variables $PM2.5$, $Precipitation$, $Humidity$, $Temperature$, GDP per capita, $Fiscal revenue$, $Health insurance$, and $Hospital$ are the annual average $PM2.5$, annual precipitation, annual average humidity, annual average temperature, GDP per capita, fiscal revenue per capita, and fraction of residents covered by basic public health insurance in city i in year t . The variable $Hospital$ is the number of public hospitals in the province to which a city belongs in year t . [3] ***, **, * denote significance at the 1%, 5% and 10% level, respectively. [4] Standard errors reported in parentheses are clustered at the city level.

Across the three disease groups, the 2SLS estimates are consistently larger than the corresponding OLS estimates, a pattern also documented in the air-pollution literature (e.g., Fu, Viard, and Zhang 2021). Because the IV estimates identify the effect of pollution variation induced by the meteorological instruments, they need not coincide with the relationship captured by the OLS estimates.

Overall, the benchmark results indicate that air pollution substantially increases drug expenditures. The responsiveness of CHM expenditures is broadly comparable to that of WM expenditures, highlighting the importance of including CHM when evaluating the pharmaceutical costs of air pollution in China.

To assess the explanatory power of PM2.5, we follow Korovkin and Makarin (2023) to compute the following statistic

$$\text{explanatory power} = \frac{[\ln(PM2.5_{p75}) - \ln(PM2.5_{p25})] \cdot |\beta_{PM2.5}|}{\ln(\text{expenditure}_{p75}) - \ln(\text{expenditure}_{p25})} \cdot 100\% \quad (2)$$

where $PM2.5_{p25}$ and $PM2.5_{p75}$ are the 25th and 75th centiles of PM2.5 values net of year and city fixed effects,⁷ $\beta_{PM2.5}$ the coefficient on PM2.5 in the 2SLS regressions, and expenditure_{p25} and expenditure_{p75} the 25th and 75th centiles of per capital expenditure net of year and fixed effects.

As summarized by Table 4, the statistics for explanatory power are 9.7%, 10.9% and 14.0% for WM drugs for treating respiratory diseases, cardiovascular diseases and tumors, respectively. The corresponding percentages for CHM drugs for the same three groups of diseases are 17.1%, 12.8% and 14.8%. Therefore, air pollution appears to be an important factor in explaining the variation in drug expenditures, especially expenditures on CHM drugs.

Our estimates suggest that a 10 $\mu\text{g}/\text{m}^3$ increase in annual average PM_{2.5} increases annual per-capita expenditure on WM and CHM drugs for the three disease groups by 108.27 yuan.⁸

⁷To obtain these values, we regress $\ln PM2.5_{i,t}$ on year and city fixed effects and obtain the residuals. The 25th and 75th centiles of the residuals are the 25th and 75th centiles of PM2.5 values net of year and city fixed effects.

⁸This figure is calculated as follows: In our sample, the mean annual satellite-measured PM2.5 is 52.3 $\mu\text{g}/\text{m}^3$ (denoted as $PM2.5_{base}$), and a 10 $\mu\text{g}/\text{m}^3$ increase corresponds to a new value of 62.3 $\mu\text{g}/\text{m}^3$ (denoted as $PM2.5_{new}$). Using the elasticity coefficients (β_g) from benchmark regressions, we calculate the predicted change in expenditure for each disease group g as $\ln(\text{expenditure}_{g,new}) - \ln(\text{expenditure}_{g,base}) = \beta_g \times [\ln(PM2.5_{new}) - \ln(PM2.5_{base})]$.

For the baseline expenditure on WM drugs treating respiratory diseases, cardiovascular diseases, and tumors (23.44, 66.21, and 122.76 yuan per capita annually from 2014 to 2019) and the respective coefficients of 2.963, 3.592, and 4.910, we use the formula: $\ln(\text{expenditure}_{g,new}) = [\ln(62.3) - \ln(52.3)] * \beta_g + \ln(\text{expenditure}_{g,base})$. Consequently, the annual expenditures for the three groups are projected to increase to 27.69, 76.55, and 153.26

Table 4: Explanatory Power of PM2.5

Variables	p25	p75	p75-p25	β	Power(%)
Panel A. Respiratory Diseases: Western medicine					
$\ln(PM2.5)$	-0.042	0.064	0.107	2.963	9.658
$\ln(expenditure\ per\ capita)$	-1.742	1.550	3.292		
Panel B. Respiratory Diseases: Chinese herbal medicine					
$\ln(PM2.5)$	-0.032	0.064	0.097	5.571	17.108
$\ln(expenditure\ per\ capita)$	-1.683	1.466	3.149		
Panel C. Cardiovascular Diseases: Western medicine					
$\ln(PM2.5)$	-0.040	0.063	0.103	3.592	10.939
$\ln(expenditure\ per\ capita)$	-1.833	1.539	3.373		
Panel D. Cardiovascular Diseases: Chinese herbal medicine					
$\ln(PM2.5)$	-0.036	0.056	0.092	4.347	12.823
$\ln(expenditure\ per\ capita)$	-1.581	1.544	3.125		
Panel E. Tumors: Western medicine					
$\ln(PM2.5)$	-0.036	0.063	0.099	4.910	14.048
$\ln(expenditure\ per\ capita)$	-1.750	1.711	3.461		
Panel F. Tumors: Chinese herbal medicine					
$\ln(PM2.5)$	-0.036	0.058	0.094	4.763	14.772
$\ln(expenditure\ per\ capita)$	-1.531	1.495	3.026		

Note: The table reports statistics for explanatory power of PM2.5, which is the fraction of interquartile range of log per capita expenditure on drugs explained by the interquartile range of log PM2.5. The formula for the statistic is $explanatory\ power = \frac{[\ln(PM2.5_{p75}) - \ln(PM2.5_{p25})] \cdot |\beta_{PM2.5}|}{\ln(expenditure_{p75}) - \ln(expenditure_{p25})} \cdot 100\%$.

This estimate is broadly consistent with the existing literature documenting the substantial medical costs of air pollution in China, although direct comparisons should be interpreted with caution because of differences in data sources, expenditure measures, and time horizons. Using nationwide credit- and debit-card transaction data, Barwick et al. (2018) estimate that a $10\ \mu g/m^3$ reduction in PM2.5 lowers annual healthcare expenditure by at least 43.47 yuan per capita. Analyzing data from the China Family Panel Studies, Liao, Du, and Chen (2021) estimate an increase of 60.84 yuan in annual per-capita medical expenditure associated with a $10\ \mu g/m^3$ increase in PM2.5. Focusing on Wuhan, Zhang et al. (2023) estimate that a $10\ \mu g/m^3$ reduction in monthly average PM2.5 lowers annual healthcare expenditure by 43.87 yuan. Taking a shorter-term perspective, Xia et al. (2022) examine short-run responses and find that a $10\ \mu g/m^3$ increase in daily PM2.5 raises healthcare expenditure by 0.38% over the yuan per capita, respectively.

For CHM drugs used in the treatment of respiratory diseases, cardiovascular diseases, and tumors, the anticipated new expenditure levels are 38.90, 124.71, and 58.54 yuan per capita annually. The total increase in annual expenditure across these six groups, 108.27 yuan per capita, is obtained by adding the new expenditures and subtracting the baseline figures: $(27.69 + 76.55 + 153.26 + 38.90 + 124.71 + 58.54) - (23.44 + 66.21 + 122.76 + 29.28 + 87.91 + 41.77)$. The amount of 108.27 yuan (about 15 US dollars at the current exchange rate) per capita annually is equivalent to 11.81% of the average annual drug expenditure in our sample.

subsequent three days.

Our estimate reinforces the conclusion that air pollution imposes substantial medical costs in China. Relative to the existing literature, our contribution is to show that CHM constitutes an economically important component of this burden. The expenditure response of CHM is broadly comparable to that of Western Medicine, implying that excluding CHM would substantially understate the pharmaceutical costs of air pollution in China.

Exploiting the information on drug quantities in our dataset, we estimate analogous regressions using quantity consumed as the dependent variable. Panel A of Table 5 reports the quantity regressions, while Panel B reproduces the benchmark expenditure results for comparison. PM2.5 is positively and significantly associated with drug quantities across all disease-medicine groups. The estimated quantity elasticities are 0.30 to 0.75 times as large as the corresponding expenditure elasticities, suggesting that increased utilization accounts for an important share of the expenditure response, while the remaining difference may reflect changes in treatment intensity and the mix of drugs consumed.

Table 5: Effects of Air Pollution on Quantity of Drugs Consumption: 2SLS

	Respiratory Diseases		Cardiovascular Diseases		Tumors	
Panel A: Dep. Var. ln(Quantity per capita)	(1) WM	(2) CHM	(3) WM	(4) CHM	(5) WM	(6) CHM
ln(PM2.5)	1.857** (0.819)	2.513** (0.893)	2.695** (1.270)	2.273** (0.956)	1.497* (0.817)	1.749* (0.783)
All other controls	Yes	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Var. Mean	1.480	1.032	1.888	1.389	0.936	1.261
N	28,638	8,748	49,110	13,249	47,205	4,033
	Respiratory Diseases		Cardiovascular Diseases		Tumors	
Panel B: Dep. Var. ln(Expenditure per capita)	(1) WM	(2) CHM	(3) WM	(4) CHM	(5) WM	(6) CHM
ln(PM2.5)	2.963** (1.142)	5.571** (2.072)	3.592** (1.628)	4.347* (2.039)	4.910* (2.402)	4.763* (2.230)
All other controls	Yes	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Var. Mean	2.423	2.957	2.617	3.952	3.623	4.667
N	28,636	8,748	49,110	13,249	47,205	4,033

Notes: [1] Table reports 2SLS estimates of equation (1) from the main text. [2] Dependent variable in Panel A is the log of quantity of drug consumption per capita of the relevant city in a year. Dependent variable in Panel B is the log of expenditure of drug consumption per capita of the relevant city in a year. [3] ***, **, * denote significance at the 1%, 5% and 10% level, respectively. [4] Standard errors reported in parenthesis are clustered at the city level. [5] All other controls include weather and socioeconomic controls.

Lastly, to examine whether the level of data aggregation affects the benchmark results,

we aggregate expenditures from the product-city-year level to the category-city-year level and re-estimate the benchmark specifications. The estimated effects are positive across all disease groups and medicine types and are broadly consistent with the benchmark results (Online Appendix Table OA1).

3.3 Significance of expenditures on CHM drugs

In this subsection, we highlight the practical significance of estimated effects of air pollution on expenditure on drugs, especially CHM drugs. In our sample, per capita expenditure on WM and CHM drugs is 652.23 yuan and 262.22 yuan, respectively.⁹ Within the per capita expenditure on WM drugs (CHM drugs, respectively), the share of WM drugs (CHM drugs) for treating respiratory diseases is 3.59% (11.17%) and the amount is 23.44 yuan (29.28 yuan).

Multiplying coefficient on PM2.5 in the 2SLS regression for WM drugs in column (2) of Table 3 with the per capita drug expenditures on WM drugs, a 1% increase in PM2.5 value is expected to increase the expenditure on WM drugs for treating respiratory diseases by 0.69 yuan ($23.44 \times 2.963\% = 0.69$ yuan), which is equivalent to 0.1% of per capita expenditure on WM drugs. The corresponding numbers for CHM drugs for treating respiratory diseases are 1.63 yuan and 0.62%.

In terms of the share in drug expenditure, cardiovascular diseases are more prominent than respiratory diseases, as the share of WM drugs (CHM drugs, respectively) for treating cardiovascular diseases make up 10.15% (33.53%) of total expenditure on all WM drugs (CHM drugs). Multiplying coefficient on PM2.5 in the 2SLS regression for WM drugs in column (6) of Table 3 with the per capita drug expenditures on WM drugs, we find that a 1% increase in PM2.5 value is associated with a 2.38 yuan rise in expenditure on WM drugs for treating cardiovascular diseases, resulting in 0.36% increase in per capital expenditure on WM drugs. The corresponding figures for CHM drugs used in the treatment of cardiovascular diseases are 3.82 yuan and 1.50%.

WM drugs (and CHM drugs, respectively) for treating tumors make up 18.82% (15.93%) of the total expenditure on all WM drugs (CHM drugs). And the expenditure on antitumor drugs is the highest among 14 groups of drugs. By multiplying the coefficient on PM2.5 in the 2SLS

⁹The per capita expenditure on WM drugs (CHM drugs, respectively) is calculated by dividing total expenditure on all WM drugs (CHM drugs) in all cities in the sample by the total population of these cities.

regression on WM drugs in column (10) of Table 3 with per capita expenditures on WM drugs, we observe that a 1% increase in the PM2.5 value is associated with a 6.03 yuan increase in expenses on WM drugs for treating tumors, resulting in a 0.92% rise in per capita expenditure on WM drugs. The corresponding figures for CHM drugs used in the treatment of tumors are 1.99 yuan and 0.76%.

Taken together, a 1% increase in PM2.5 causes practically significant rise in drug expenditures. The combined increase in drug expenditures on WM drugs at 9.10 yuan (the sum of 0.69 yuan for respiratory diseases, 2.38 yuan for cardiovascular diseases, and 6.03 for tumors) which is equal to 1.00% of the per capital expenditure on all drugs in our sample. In comparison, following a 1% rise in PM2.5, the combined increase in drug expenditures on CHM drugs at 7.44 yuan (the sum of 1.63 yuan for respiratory diseases, 3.82 yuan for cardiovascular diseases, and 1.99 for tumors) which is equal to 0.81% of the per capital expenditure on all drugs in our sample. To the best of our knowledge, we are the first to examine separately expenditures on WM drugs and CHM drugs. Our results highlight the importance of CHM drugs in the understanding of drug expenditures in general.

4 Regional Disparities in Response of Drug Expenditures to Pollution

4.1 Socioeconomic factors

Existing research indicates that individual-level factors affect their exposure to pollution. Individuals at low socioeconomic status, measured by income, level of education, occupation, ethnicity, nationality, social security enrollment, neighborhood, and household registration status, are disproportionately exposed to environmental pollution (Tonne et al., 2008; Currie, Neidell, and Schmieder, 2009; Currie, 2011). Recognizing the importance of socioeconomic factors, we examine whether the regional disparities in three such factors contribute to the disparities in drug expenditures associated with air pollution.

The first factor is local fiscal resources. Differences in fiscal resources across countries are known to be a driver of disparities in health care spending (WHO et al., 2018). In the context of China, because a significant portion of local public health programs is financed by municipal governments, the generosity of public medical insurance differs across cities.¹⁰ The second

¹⁰The State Council acknowledged substantial differences in the coverage of local public medical insurance

factor is household disposable income. If the burden of drug costs is significant enough, we should observe that cities with lower per capita disposable income will spend less on drugs following a rise in air pollution. The third factor is education attainment. Individuals with better education are more likely to possess better knowledge of air pollution and seek medical help after observing symptoms related to pollution.

To capture the moderating effects of these three socioeconomic factors, we introduce the interactions of fiscal revenue per resident, per capita disposable income, the fraction of college graduates among residents with PM_{2.5} into the benchmark regressions.¹¹ Due to our focus on these socioeconomic factors, in this section, we pool the observations from all three disease groups to estimate the interaction effects. In the regressions reported in Table 6, we remove the sample mean from all variables that are interacted, such that the main coefficient on each of these variables is the marginal effect of the variable evaluated at the sample mean.

The interaction terms between PM_{2.5} and all three socioeconomic characteristics are positive for both WM and CHM expenditures, suggesting larger expenditure responses to air pollution in cities with greater fiscal resources, higher household income, and a larger share of college graduates. However, statistical significance is concentrated in the WM regressions. The interaction between household income and PM_{2.5} is 1.892 and statistically significant at the 10% level, while the interaction between education and PM_{2.5} is 0.112 and significant at the 5% level. The corresponding interactions for CHM expenditures are positive but statistically insignificant. The fiscal-revenue interactions are also positive for both WM and CHM expenditures but are not statistically significant.

The magnitudes of the estimated interactions for WM are economically meaningful. A one-standard-deviation increase in log household disposable income (0.275) is associated with an increase of 0.52 in the pollution elasticity of WM expenditure. Similarly, for cities in which the

programs in its policy document Key Tasks for Further Reform of the Medical and Health Care System in 2022 (*Shenhua Yiyao Weisheng Tizhi Gaige 2022 Nian Zhongdian Gongzuo Renwu*) and called for further integration of these programs across provinces. To examine whether insurance generosity affects the expenditure response to air pollution, we collect city-level employee inpatient reimbursement rates from local policy documents. Because reimbursement rates exhibit little within-city variation during our sample period and are therefore largely absorbed by city fixed effects, we interact them with time-varying PM_{2.5}. Online Appendix Table OA3 shows that the interaction between $\ln(\text{PM}_{2.5})$ and the reimbursement rate is positive and statistically significant for WM expenditures, while the corresponding estimate for CHM expenditures is positive but statistically insignificant. These results provide supporting evidence that greater insurance generosity is associated with a larger pharmaceutical expenditure response to air pollution.

¹¹The three variables are retrieved from the National Bureau of Statistics of China and China Health Statistical Yearbook.

Table 6: Socioeconomic Factors and Regional Disparities in Drug Expenditure

	Western medicine:2SLS			Chinese herbal medicine:2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)
ln(PM2.5)	4.038** (1.540)	4.209* (2.065)	6.197** (2.251)	5.290** (1.913)	5.762** (2.396)	7.157*** (1.917)
ln(Fiscal revenue)	-2.742 (24.182)	4.693 (4.903)	6.760 (6.506)	-12.547 (17.208)	-1.838 (6.047)	3.172 (5.955)
ln(PM2.5)*ln(Fiscal revenue)	1.430 (6.236)			3.650 (4.155)		
ln(HHD income)		-6.277 (4.401)			-5.450 (6.037)	
ln(PM2.5)*ln(HHD income)		1.892* (1.062)			1.861 (1.679)	
Education			-0.376* (0.184)			-0.074 (0.089)
ln(PM2.5)*Education			0.112** (0.049)			0.008 (0.022)
All other controls	Yes	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Category FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Var. Mean	2.952	2.952	2.952	3.728	3.728	3.728
CD F statistic	2810.659	2350.832	2473.170	913.494	950.479	1354.324
KP F statistic	3.166	5.809	7.553	4.604	12.667	11.174
N	124,951	124,951	124,951	26,030	26,030	26,030

Notes: [1] Table reports heterogeneity in pollution-induced drug expenditure responses by socioeconomic factors. The dependent variable is the log of per capita drug expenditure. [2] ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. [3] Standard errors are clustered at the city level. [4] All other controls include socioeconomic and weather controls.

share of college graduates is one standard deviation (8.473 percentage points) above the sample mean, the estimated pollution elasticity of WM expenditure is higher by 0.95.

Overall, the point estimates consistently indicate larger pharmaceutical expenditure responses to air pollution in cities with greater fiscal resources, higher household income, and a larger share of college graduates. The statistical evidence, however, is concentrated in WM expenditure: the education interaction is significant at the 5% level and the household-income interaction at the 10% level, whereas the fiscal-revenue interaction and all three interactions for CHM are statistically insignificant. We therefore interpret these results as evidence that WM expenditures respond more strongly to air pollution in cities with a larger share of college graduates, with suggestive evidence of a similar pattern in cities with higher household income.

4.2 Northern vs. Southern Chinese Cities

Disparities in health care utilization can be ascribed not only to socioeconomic factors detailed in the preceding subsection, but also to the geographical region of residence (Sözmen and Ünal, 2016). When it comes to China, air quality in northern China is poorer than southern China because of heavier use of coal in heating seasons, less precipitation, and other factors. The literature on air pollution in China has established that the poor air quality in northern China lowers life expectancy and raises mortality rate (Chen et al., 2013; Ebenstein et al., 2017; Gong et al., 2023). Therefore, residents in northern China might be forced to spend more to treat pollution-related diseases. On the other hand, because humans can adapt to the environment (Komolafe et al., 2014), expenditure effects of air pollution in the north may be mitigated by such adaptation.

To investigate whether the poor air quality in northern China leads to an expenditure pattern different from southern China, we introduce to 2SLS regressions an indicator variable for northern cities and interact it with air pollution.¹² Similar to Subsection 4.1, we pool all observations from all three diseases groups. As presented in Table 7, the coefficients on the interaction term between PM2.5 and indicator for northern cities are positive and significant, which indicates a larger response of drug expenditure to air pollution in northern cities than southern cities. As we alternate the dependent variable among expenditure on WM and CHM

¹²In our samples, the northern cities include Beijing, Changchun, Harbin, Jinan, Shijiazhuang, Shenyang, Taiyuan, Tianjin, Xi'an, and Zhengzhou; the southern cities include Changsha, Chengdu, Chongqing, Fuzhou, Guangzhou, Hangzhou, Nanjing, Shanghai, Shenzhen, and Wuhan.

drugs, the coefficients on the interaction term are 1.02 and 2.03. Overall, the difference in pollution elasticities between southern and northern China is significant in both economic and statistical sense.

Table 7: Difference between Northern and Southern Cities of China

	(1)	(2)
	Western medicine: 2SLS	Chinese herbal medicine: 2SLS
$\ln(\text{PM}_{2.5}) \times \text{North}$	1.022* (0.567)	2.029** (0.666)
$\ln(\text{PM}_{2.5})$	3.573* (1.707)	3.678 (2.354)
All other controls	Yes	Yes
City FE	Yes	Yes
Year FE	Yes	Yes
Category FE	Yes	Yes
Dep. Var. Mean	2.952	3.728
CD F statistic	2764.535	1140.544
KP F statistic	3.635	3.428
N	124,951	26,030

Notes: [1] Table reports disparities between northern and southern Chinese cities. [2] Dependent variable is the log of expenditure on drug per capita of the relevant city in a year. [3] ***, ** and * are significance levels at 1%, 5% and 10%, respectively. [4] Standard errors, clustered at the city level, are reported in parenthesis. [5] All other controls include weather and socio-economic controls.

The findings in Table 7 suggest that northern residents' drug expenditure is significantly more responsive to rise in air pollution than their southern counterparts. In particular, the north-south difference in sensitivity of drug expenditure is larger for CHM drugs than WM drugs. Based on these results, targeted interventions aimed at reducing air pollution in the northern cities may be particularly effective in mitigating the burden of drug expenditure.

5 Robustness and Extensions

5.1 Robustness checks

To verify that the benchmark findings are not sensitive to the measure of air pollution, we consider several alternative measures. First, to alleviate concerns that the satellite-based TAP $\text{PM}_{2.5}$ measure may be subject to measurement error, we use $\text{PM}_{2.5}$ from the Air Quality Index (AQI) published by the Chinese government, which is based on readings from ground monitoring stations. Second, because short episodes of severe pollution may have disproportionate health effects, we replace annual average $\text{PM}_{2.5}$ with the annual maximum $\text{PM}_{2.5}$ using both the TAP and AQI datasets. For ease of comparison, column (1) of Panels A and B in Table 8

reproduces the benchmark regressions for respiratory diseases, while the remaining columns report estimates using the alternative pollution measures.

Table 8: Alternative Measures of Air Pollution across Disease Categories

Panel A:	Respiratory Diseases				Cardiovascular Diseases				Tumors			
Western medicine	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
ln(PM2.5)	2.963** (1.142)				3.592** (1.628)				4.910* (2.402)			
ln(PM2.5 max)		1.906** (0.715)				2.065** (0.861)				2.362** (1.062)		
ln(AQI:PM2.5 mean)			2.296** (1.071)				3.041* (1.651)				4.757 (3.209)	
ln(AQI:PM2.5 max)				1.552*** (0.415)				1.708*** (0.500)				1.980*** (0.579)
All other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Var. Mean	2.423	2.423	2.455	2.455	2.617	2.617	2.645	2.645	3.623	3.623	3.651	3.651
N	28,636	28,636	28,215	28,215	49,110	49,110	48,457	48,457	47,205	47,205	46,675	46,675
Panel B:	Respiratory Diseases				Cardiovascular Diseases				Tumors			
Chinese herbal medicine	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
ln(PM2.5)	5.571** (2.072)				4.347* (2.039)				4.763* (2.230)			
ln(PM2.5 max)		3.934** (1.303)				2.788* (1.220)				3.239** (1.266)		
ln(AQI:PM2.5 mean)			12.003 (12.240)				2.882 (3.250)				2.907 (5.082)	
ln(AQI:PM2.5 max)				2.919** (0.875)				2.277*** (0.651)				2.537** (0.827)
All other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Var. Mean	2.957	2.957	2.996	2.996	3.952	3.952	3.982	3.982	4.667	4.667	4.698	4.698
N	8,748	8,748	8,587	8,587	13,249	13,249	13,103	13,103	4,033	4,033	3,995	3,995

Notes: [1] Table reports 2SLS estimates of equation (1) from the main text. [2] ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. [3] Standard errors are clustered at the city level. [4] All other controls include socioeconomic and weather controls.

The point estimates remain positive across all alternative PM_{2.5} measures. Estimates based on annual maximum PM_{2.5} from either TAP or AQI data are statistically significant at the 10% level or better for all six disease–medicine groups. With AQI-based annual mean PM_{2.5}, the WM estimates are significant at the 5% level for respiratory drugs and at the 10% level for cardiovascular drugs, whereas the estimate for WM antitumor drugs and all three CHM estimates are not statistically significant. Thus, the direction of the response is consistent across pollution measures, but statistical significance is not uniform.

We next conduct additional analyses addressing two potential concerns about identification. First, to examine whether the IV results depend on the particular set of meteorological instruments, we re-estimate the benchmark specifications using temperature inversion alone and other alternative IV specifications. The results remain consistent with the benchmark estimates (Online Appendix Section G). Second, to assess the potential importance of migration-related selection, we use available migration-flow data for 2015 and examine whether air pollution is sys-

tematically associated with migration flows. We find little evidence that more polluted regions experienced greater out-migration, suggesting that migration is unlikely to be a quantitatively important source of bias in our setting (Online Appendix Section H).

Additional specifications examine clustering by city and year, exclusion of extreme-pollution city-years, and population weighting (Online Appendix Tables OA4–OA6). The 2SLS point estimates in these specifications remain positive and significant across the six disease–medicine groups.

5.2 Persistence of pollution effects

To examine whether the impact of air pollution persists beyond one year, we augment the benchmark specification by including a one-year lag of PM2.5.¹³ The benchmark estimates for respiratory diseases, cardiovascular diseases, and tumors are reproduced in columns (1), (3), and (5), respectively, while columns (2), (4), and (6) report the corresponding regressions including lagged pollution.

The results indicate that the effects of air pollution on WM drug expenditures persist beyond one year, while the evidence for CHM is more mixed. For WM drugs, lagged PM2.5 has a positive and statistically significant effect on expenditures for all three disease groups, although the contemporaneous effect becomes insignificant for respiratory diseases. For CHM drugs, both contemporaneous and lagged PM2.5 are significant for cardiovascular diseases, whereas only the contemporaneous effect is significant for respiratory diseases and neither is significant for tumors. Overall, the persistence of pollution-related expenditure responses is more pronounced for WM than for CHM drugs.

¹³When further lags are introduced in unreported regressions, they are statistically insignificant.

Table 9: Longer-term Effects of Air Pollution

Panel A:	Respiratory Diseases		Cardiovascular Diseases		Tumors	
Western medicine	(1)	(2)	(3)	(4)	(5)	(6)
ln(PM2.5)	2.963** (1.142)	0.370 (1.091)	3.592** (1.628)	0.483 (1.174)	4.910* (2.402)	2.240 (1.659)
Lag.ln(PM2.5)		0.531* (0.288)		0.947*** (0.308)		1.055** (0.402)
All other controls	Yes	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Var. Mean	2.423	2.678	2.617	2.871	3.623	3.873
N	28,636	20,605	49,110	35,657	47,205	34,539
Panel B:	Respiratory Diseases		Cardiovascular Diseases		Tumors	
Chinese herbal medicine	(1)	(2)	(3)	(4)	(5)	(6)
ln(PM2.5)	5.571** (2.072)	4.076 (2.604)	4.347* (2.039)	1.956 (1.667)	4.763* (2.230)	1.814 (1.269)
Lag.ln(PM2.5)		-0.058 (0.483)		0.737 (0.525)		0.915* (0.485)
All other controls	Yes	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Var. Mean	2.957	3.184	3.952	4.165	4.667	4.886
N	8,748	6,224	13,249	9,901	4,033	3,057

Notes: [1] Table reports 2SLS estimates of equation (1) augmented with a one-year lag of PM2.5. [2] ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. [3] Standard errors are clustered at the city level. [4] All other controls include socioeconomic and weather controls.

6 Conclusion

Using panel data on drug expenditures from major public hospitals in 20 large Chinese cities from 2014 to 2019, we estimate how air pollution affects expenditures on Western Medicine (WM) and Chinese Herbal Medicine (CHM) drugs used to treat respiratory diseases, cardiovascular diseases, and tumors. To address potential endogeneity arising from omitted variables, we employ meteorological instruments for PM_{2.5}.

Our results indicate that air pollution imposes a substantial economic burden through higher pharmaceutical expenditures. A 1% increase in annual average PM_{2.5} increases spending on drugs used to treat the three disease groups by an amount equivalent to 1.81% of annual per-capita drug expenditure. In absolute terms, our estimates imply that a 10 $\mu\text{g}/\text{m}^3$ increase in annual average PM_{2.5} increases annual per-capita drug expenditure by approximately 108 yuan (US\$15).

The responsiveness of CHM expenditures is broadly comparable to that of WM expenditures, indicating that CHM constitutes an important component of the pharmaceutical costs of air pollution in China. We also find evidence of socioeconomic heterogeneity in WM expenditures: the expenditure response is significantly larger in cities with a higher share of college graduates, with suggestive evidence of a larger response in cities with higher household income. Finally, pollution-related increases in drug expenditures are more pronounced in Northern than in Southern China.

CRedit authorship contribution statement

Heng Ju: Conceptualization, Resources, Data Curation, Writing - Review & Editing, Supervision, Funding acquisition. **Yao Tang:** Conceptualization, Methodology, Validation, Formal analysis, Writing - Review & Editing, Project administration, Funding acquisition. **Meilan Zhang:** Conceptualization, Software, Formal analysis, Investigation, Writing - Original Draft, Writing - Review & Editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used GPT5.2 in order to improve readability and language. After using this service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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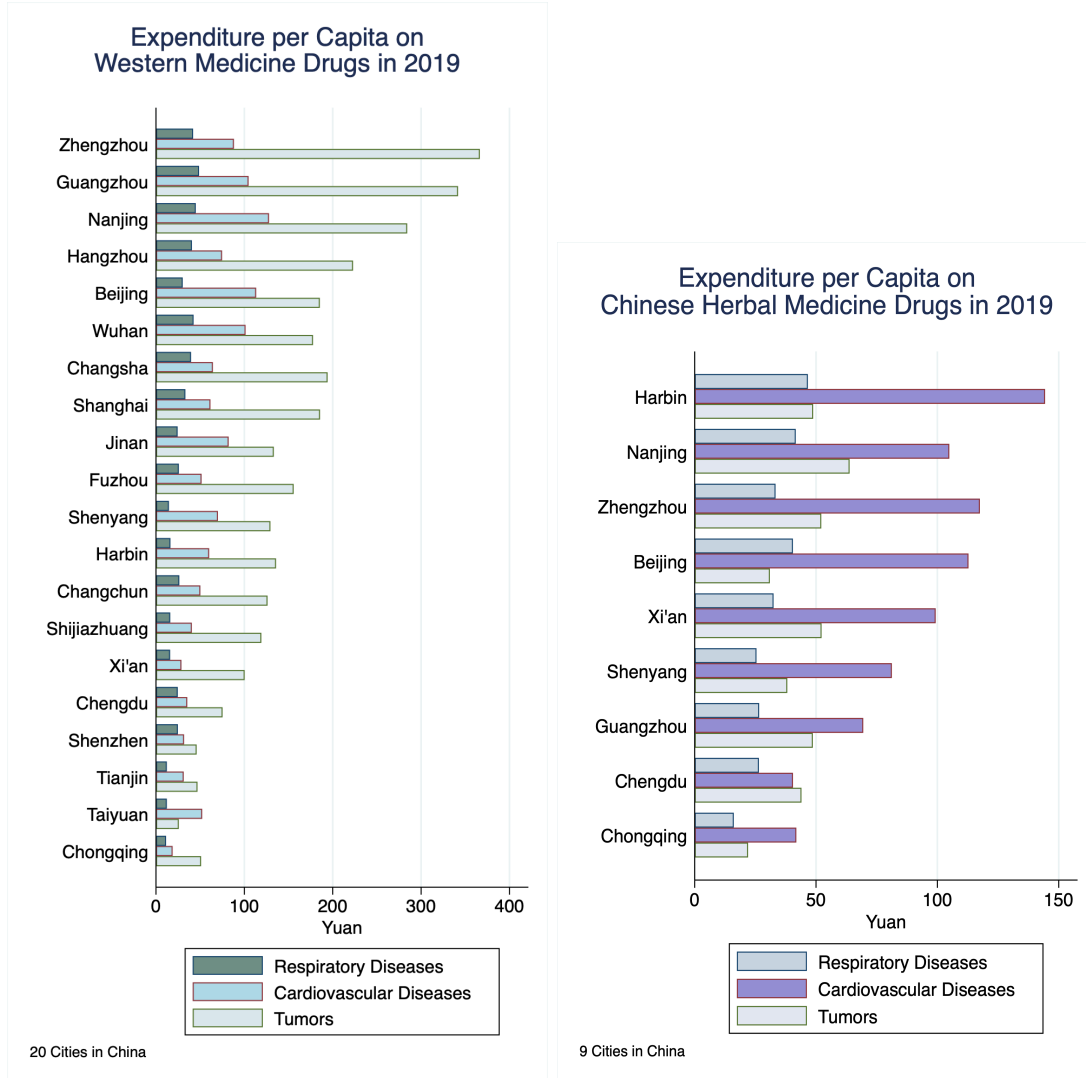
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Online Appendix

A. City-level Expenditures on WM drugs and CHM drugs

Figure OA1: City-level Expenditures on WM drugs and CHM drugs



Notes : The left panel shows the current price expenditures per capita on Western medicine drugs for treating respiratory diseases, cardiovascular diseases and tumors in 20 cities in 2019. The right panel shows the current price expenditures per capita on Chinese herbal medicine drugs for treating respiratory diseases, cardiovascular diseases and tumors in nine cities in 2019. Source: authors' calculation.

B. Robustness to Expenditure Aggregation

Table OA1: Drug Expenditure Aggregated to the City-Year-Category Level

	Respiratory Diseases				Cardiocerebral Vascular Diseases				Tumors			
	(1) WM OLS	(2) WM 2SLS	(3) CHM OLS	(4) CHM 2SLS	(5) WM OLS	(6) WM 2SLS	(7) CHM OLS	(8) CHM 2SLS	(9) WM OLS	(10) WM 2SLS	(11) CHM OLS	(12) CHM 2SLS
ln(PM2.5)	0.656*** (0.229)	3.860** (1.699)	1.344*** (0.155)	6.168** (2.307)	0.663*** (0.234)	3.299 (1.965)	1.355*** (0.241)	5.398* (2.724)	0.997*** (0.314)	4.745* (2.585)	1.479*** (0.258)	5.352** (2.118)
ln(Precipitation)	0.093 (0.069)	0.564** (0.265)	0.331*** (0.128)	1.540** (0.528)	0.166*** (0.058)	0.532** (0.226)	0.371** (0.178)	1.158 (0.650)	0.121 (0.099)	0.539* (0.288)	0.151 (0.155)	0.746 (0.458)
ln(Humidity)	-0.046 (0.255)	0.467 (0.732)	-0.440 (0.288)	-1.241 (0.934)	-0.244 (0.276)	0.030 (0.699)	-0.934** (0.364)	-1.535 (1.346)	0.091 (0.356)	0.235 (0.740)	-0.902** (0.441)	-1.318 (1.403)
ln(Temperature)	0.365** (0.167)	3.492** (1.596)	0.835*** (0.148)	4.364** (1.754)	0.330* (0.197)	2.654 (1.628)	0.909*** (0.182)	4.215* (2.010)	0.675*** (0.237)	4.073* (2.311)	1.188*** (0.224)	3.862** (1.555)
ln(GDP per capita)	2.095*** (0.295)	5.244*** (0.530)	2.311*** (0.383)	5.234*** (0.768)	1.802*** (0.292)	5.060*** (0.687)	2.442*** (0.435)	5.912*** (0.849)	2.816*** (0.490)	5.507*** (0.810)	2.849*** (0.710)	6.785*** (1.114)
ln(Fiscal revenue)	-0.925 (1.895)	0.312 (5.154)	1.600 (1.938)	0.922 (6.677)	0.204 (1.697)	1.084 (4.232)	-0.305 (2.185)	-1.398 (6.924)	0.899 (1.909)	2.693 (5.502)	2.532 (2.918)	0.783 (6.222)
ln(Insurance)	0.186*** (0.059)	0.701*** (0.199)	0.357*** (0.049)	1.092*** (0.248)	0.186*** (0.051)	0.615*** (0.186)	0.367*** (0.057)	0.977*** (0.260)	0.253*** (0.053)	0.717*** (0.217)	0.547*** (0.085)	1.183*** (0.254)
ln(Hospital)	1.025** (0.444)	1.584 (1.270)	-0.046 (0.428)	-0.734 (0.934)	0.974** (0.387)	1.688 (1.015)	0.522 (0.555)	-0.073 (1.149)	1.375*** (0.403)	1.666 (1.288)	0.826 (0.818)	-0.215 (0.776)
Constant	-8.676** (4.363)		-4.560 (3.314)		-7.032* (3.890)		-8.600** (3.814)		-13.274*** (4.238)		-9.222 (6.103)	
City FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Category2 FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Var. Mean	3.196	3.196	3.527	3.527	3.704	3.704	4.432	4.432	4.630	4.630	5.220	5.220
CD F statistic		509.951		341.061		834.234		500.255		769.476		190.483
KP F statistic		5.938		4.958		7.531		5.640		6.537		7.319
Within R ²	0.123		0.174		0.104		0.154		0.160		0.165	
N	13,148	13,148	5,761	5,761	19,596	19,596	8,381	8,381	20,238	20,238	2,619	2,619

Notes: Medicine expenditures are aggregated to the category-city-year level, and the dependent variable is $\ln(1 + \text{total expenditure})$. A drug category is defined as a distinct $\text{Category2} \times \text{PriceTier} \times \text{ProductFamily}$ combination. *Category2* denotes the second-level therapeutic classification; *PriceTier* is based on quintiles of each product's baseline price, measured by its median price in the first year in which the product is observed; and *ProductFamily* groups normalized product names after removing dosage-form and administration-related descriptors. The number of resulting categories is reported in Table OA2. WM denotes Western medicine and CHM denotes Chinese herbal medicine. Odd-numbered columns report OLS estimates and even-numbered columns report 2SLS estimates. All specifications include city, year, and category fixed effects and otherwise follow the baseline regressions in terms of control variables, sample restrictions, winsorization, and the instrumental-variable strategy. Standard errors are clustered at the city level and reported in parentheses. CD and KP F statistics report the corresponding first-stage instrument-strength tests. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table OA2: The Number of Drug Categories in Regressions in Table OA1

Disease group	Medicine	# of Drugs	# of Drug Categories
Respiratory diseases	WM	13,148	292
Respiratory diseases	CHM	5,761	445
Cardiocerebral vascular diseases	WM	19,596	317
Cardiocerebral vascular diseases	CHM	8,381	424
Tumors	WM	20,238	300
Tumors	CHM	2,619	109

Notes: WM denotes Western medicine and CHM denotes Chinese herbal medicine.

C. Heterogeneity by Insurance Reimbursement Rates

Table OA3: Drug Expenditure Responses by Public Medical Insurance Reimbursement Rates

	Modern: 2SLS (1)	Traditional: 2SLS (2)
Dependent variable: $\ln(\text{Drug expenditure per capita})$		
$\ln(\text{PM}_{2.5})$	-5.041** (2.288)	0.376 (3.228)
$\ln(\text{PM}_{2.5}) \times \text{reimbursement rate}$	8.246*** (2.536)	4.648 (3.526)
All other controls	Yes	Yes
City FE	Yes	Yes
Year FE	Yes	Yes
Category FE	Yes	Yes
Dep. Var. Mean	2.947	3.728
CD F statistic	555.590	1004.475
KP F statistic	75.268	357.811
N	118,132	26,030

Notes: This table reports 2SLS estimates of heterogeneity in the effect of air pollution on drug expenditure by the hospital reimbursement rate. The dependent variable is the log of annual per capita expenditure on drug j in city i and year t . The hospital reimbursement rate is the city-level employee inpatient reimbursement ratio for inpatient care and is measured on a scale from 0 to 1. Because the reimbursement rate exhibits no within-city variation in the estimation sample, its main effect is absorbed by city fixed effects. Both $\ln(\text{PM}_{2.5})$ and its interaction with the reimbursement rate are treated as endogenous and instrumented using the baseline meteorological instruments and their interactions with the reimbursement rate. All specifications include the benchmark weather and socioeconomic controls, as well as city, year, and category fixed effects. Standard errors in parentheses are clustered at the city level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

D. Robustness to Alternative Clustering

Table OA4: Benchmark Regressions with Two-Way Clustering by City and Year

	Respiratory Diseases				Cardiocerebral Vascular Diseases				Tumors			
	(1) WM OLS	(2) WM 2SLS	(3) CHM OLS	(4) CHM 2SLS	(5) WM OLS	(6) WM 2SLS	(7) CHM OLS	(8) CHM 2SLS	(9) WM OLS	(10) WM 2SLS	(11) CHM OLS	(12) CHM 2SLS
ln(PM2.5)	0.952*** (0.223)	2.963** (1.304)	1.623*** (0.401)	5.571** (2.283)	0.829*** (0.240)	3.592** (1.636)	1.998*** (0.439)	4.347** (2.097)	1.268*** (0.332)	4.910** (2.240)	1.928*** (0.446)	4.763** (1.920)
ln(Precipitation)	0.196** (0.094)	0.522** (0.209)	0.415** (0.168)	1.319*** (0.468)	0.204** (0.098)	0.587** (0.245)	0.416** (0.208)	1.098** (0.522)	0.274** (0.131)	0.701** (0.300)	0.567*** (0.220)	1.082** (0.482)
ln(Humidity)	-0.181 (0.258)	0.288 (0.487)	-0.274 (0.440)	-0.700 (0.974)	-0.145 (0.260)	-0.066 (0.533)	-0.798 (0.531)	-0.586 (1.028)	-0.217 (0.385)	0.033 (0.691)	-0.712 (0.517)	-1.262 (1.015)
ln(Temperature)	0.516*** (0.158)	2.536** (1.045)	0.894*** (0.253)	4.176** (1.581)	0.573*** (0.168)	2.905** (1.274)	1.284*** (0.289)	3.256** (1.366)	0.863*** (0.231)	4.050** (1.795)	1.072*** (0.298)	3.516** (1.352)
ln(GDP per capita)	2.638*** (0.294)	5.035*** (0.427)	2.713*** (0.401)	4.972*** (0.571)	2.696*** (0.316)	5.329*** (0.497)	3.260*** (0.548)	6.414*** (0.805)	3.506*** (0.455)	6.047*** (0.676)	3.707*** (0.680)	6.929*** (0.887)
ln(Fiscal revenue)	0.282 (1.409)	1.252 (2.995)	0.830 (2.123)	2.081 (5.074)	-0.316 (1.491)	1.493 (3.383)	-1.886 (2.450)	-1.355 (4.826)	1.473 (1.798)	3.542 (4.240)	2.274 (2.708)	1.269 (5.359)
ln(Insurance)	0.216** (0.084)	0.611*** (0.214)	0.320*** (0.116)	0.946*** (0.340)	0.248*** (0.083)	0.668*** (0.221)	0.490*** (0.136)	1.010*** (0.317)	0.379*** (0.106)	0.826*** (0.286)	0.669*** (0.137)	1.210*** (0.306)
ln(Hospital)	1.161** (0.470)	1.518 (0.931)	0.121 (0.627)	-0.921 (1.537)	1.138** (0.455)	1.501 (0.948)	0.331 (0.679)	0.455 (1.352)	1.572*** (0.600)	2.051* (1.170)	1.392* (0.733)	1.076 (1.389)
Constant	-12.243*** (4.032)		-8.514** (4.340)		-11.624*** (3.863)		-10.196* (5.506)		-16.849*** (5.238)		-16.769*** (5.401)	
City FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Var. Mean	2.423	2.423	2.957	2.957	2.617	2.617	3.952	3.952	3.623	3.623	4.667	4.667
CD F statistic		1270.697		558.707		1995.189		941.627		1842.036		277.304
KP F statistic		13.278		9.204		12.304		10.455		11.911		8.586
Within R ²	0.224		0.268		0.198		0.267		0.247		0.265	
N	28,636	28,636	8,748	8,748	49,110	49,110	13,249	13,249	47,205	47,205	4,033	4,033

Notes: This table re-estimates the benchmark regressions using two-way clustering by city and year. All other specifications remain unchanged. Standard errors are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

E. Robustness to Excluding Extreme Pollution Observations

Table OA5: Benchmark Regressions Excluding Extreme Pollution City-Years

	Respiratory Tract Diseases				Cardiocerebral Vascular Diseases				Tumors			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	WM OLS	WM 2SLS	CHM OLS	CHM 2SLS	WM OLS	WM 2SLS	CHM OLS	CHM 2SLS	WM OLS	WM 2SLS	CHM OLS	CHM 2SLS
ln(PM2.5)	1.063*** (0.271)	2.627** (1.146)	1.633*** (0.405)	5.506* (2.428)	0.932*** (0.282)	3.021* (1.582)	2.034*** (0.410)	4.586* (2.432)	1.384*** (0.395)	4.618* (2.402)	1.931*** (0.537)	5.643 (3.312)
ln(Precipitation)	0.188** (0.087)	0.464** (0.210)	0.316** (0.127)	1.195** (0.476)	0.200** (0.084)	0.528* (0.264)	0.317* (0.164)	1.009 (0.589)	0.285** (0.125)	0.693* (0.354)	0.506** (0.239)	1.064* (0.540)
ln(Humidity)	-0.155 (0.263)	0.347 (0.648)	-0.332 (0.612)	-1.015 (1.433)	-0.221 (0.291)	-0.028 (0.601)	-0.948 (0.725)	-1.124 (1.535)	-0.347 (0.398)	-0.078 (0.853)	-0.963 (0.660)	-1.767 (1.670)
ln(Temperature)	0.569*** (0.197)	2.206** (0.989)	0.985*** (0.264)	4.103* (1.786)	0.655*** (0.198)	2.376* (1.220)	1.443*** (0.293)	3.402* (1.653)	0.964*** (0.254)	3.618* (1.873)	1.122*** (0.425)	4.052* (2.173)
ln(GDP per capita)	2.598*** (0.341)	5.035*** (0.443)	2.845*** (0.422)	5.284*** (0.704)	2.634*** (0.351)	5.366*** (0.539)	3.387*** (0.594)	6.620*** (0.897)	3.379*** (0.511)	6.019*** (0.776)	3.849*** (0.772)	6.982*** (1.036)
ln(Fiscal revenue)	0.689 (1.814)	1.169 (3.975)	-0.726 (2.103)	1.787 (6.015)	0.123 (1.804)	1.642 (4.526)	-2.720 (2.113)	-1.585 (5.407)	1.927 (2.260)	4.083 (6.097)	0.737 (3.206)	2.402 (7.419)
ln(Insurance)	0.237*** (0.054)	0.626*** (0.152)	0.314*** (0.069)	1.005*** (0.265)	0.268*** (0.058)	0.693*** (0.186)	0.486*** (0.056)	1.105*** (0.240)	0.388*** (0.058)	0.859*** (0.228)	0.644*** (0.054)	1.326*** (0.319)
ln(Hospital)	1.200** (0.491)	1.507 (1.120)	-0.176 (0.515)	-0.832 (1.416)	1.151** (0.521)	1.584 (1.095)	0.223 (0.451)	0.469 (1.053)	1.588*** (0.584)	2.103 (1.396)	1.145 (0.720)	1.177 (1.256)
Constant	-13.223*** (4.903)		-6.508* (3.595)		-12.089** (5.034)		-9.420*** (3.314)		-17.212*** (5.786)		-14.192*** (4.410)	
PM2.5 Support	P2.5-P97.5	P2.5-P97.5	P2.5-P97.5	P2.5-P97.5	P2.5-P97.5	P2.5-P97.5	P2.5-P97.5	P2.5-P97.5	P2.5-P97.5	P2.5-P97.5	P2.5-P97.5	P2.5-P97.5
City FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Var. Mean	2.474	2.474	3.046	3.046	2.659	2.659	4.031	4.031	3.660	3.660	4.754	4.754
CD F statistic		1234.250		433.826		1810.035		691.098		1646.117		204.432
KP F statistic		5.473		3.851		4.914		3.468		4.357		2.910
Within R ²	0.218		0.267		0.193		0.271		0.235		0.262	
N	26,869	26,869	7,931	7,931	46,202	46,202	12,152	12,152	44,492	44,492	3,728	3,728

Notes: This table re-estimates the benchmark OLS and 2SLS regressions after excluding city-year observations with annual PM_{2.5} concentrations below the 2.5th or above the 97.5th percentile of the sample distribution. All other specifications remain unchanged. Standard errors are clustered at the city level and reported in parentheses. CD and KP F statistics report the corresponding first-stage instrument-strength tests. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

F. Robustness to Population Weighting

Table OA6: Population-Weighted Benchmark Regressions

	Respiratory Diseases				Cardiocerebral Vascular Diseases				Tumors			
	(1) WM OLS	(2) WM 2SLS	(3) CHM OLS	(4) CHM 2SLS	(5) WM OLS	(6) WM 2SLS	(7) CHM OLS	(8) CHM 2SLS	(9) WM OLS	(10) WM 2SLS	(11) CHM OLS	(12) CHM 2SLS
ln(PM2.5)	1.426** (0.547)	4.109** (1.644)	2.744** (0.862)	7.997** (2.960)	1.647** (0.647)	4.886** (2.272)	3.124** (1.094)	7.706** (2.959)	2.217** (0.822)	6.260* (3.050)	3.205** (1.316)	7.844 (4.438)
ln(Precipitation)	0.404* (0.193)	0.687** (0.282)	0.857** (0.371)	1.696** (0.569)	0.421** (0.194)	0.773** (0.339)	1.012* (0.530)	1.725** (0.744)	0.575** (0.246)	0.950** (0.437)	1.061* (0.464)	1.637* (0.725)
ln(Humidity)	0.618 (0.509)	0.628 (0.605)	0.508 (0.653)	0.131 (1.309)	0.432 (0.454)	0.337 (0.607)	0.627 (0.703)	0.502 (1.239)	0.440 (0.484)	0.414 (0.663)	-0.239 (0.739)	0.044 (1.101)
ln(Temperature)	1.410** (0.492)	3.605** (1.619)	2.072*** (0.426)	6.101* (2.700)	1.434*** (0.495)	4.099* (2.057)	2.188*** (0.494)	5.535* (2.530)	1.860*** (0.607)	5.303* (2.779)	2.249*** (0.580)	5.869 (3.606)
ln(GDP per capita)	5.116*** (0.423)	4.811*** (0.433)	5.164*** (0.510)	4.648*** (0.446)	5.362*** (0.464)	4.984*** (0.543)	6.352*** (0.654)	5.937*** (0.599)	6.159*** (0.622)	5.681*** (0.744)	7.014*** (0.637)	6.637*** (0.672)
ln(Fiscal revenue)	1.939 (3.198)	4.566 (4.416)	0.722 (4.877)	5.582 (8.434)	1.254 (3.448)	4.721 (5.256)	0.678 (5.698)	4.037 (8.142)	3.168 (4.304)	6.834 (6.298)	4.812 (6.108)	8.200 (9.538)
ln(Insurance)	0.516*** (0.138)	0.701*** (0.214)	0.701** (0.228)	1.042** (0.368)	0.535*** (0.144)	0.750*** (0.256)	0.914*** (0.208)	1.221*** (0.299)	0.701*** (0.163)	0.915*** (0.274)	1.142*** (0.245)	1.477*** (0.408)
ln(Hospital)	2.608** (0.993)	2.638** (1.137)	-0.708 (0.943)	-2.102 (1.267)	2.566** (0.952)	2.599** (1.179)	0.509 (1.084)	-0.104 (1.487)	3.417*** (1.134)	3.380** (1.447)	0.999 (1.368)	0.470 (1.286)
City FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Population Weight	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Var. Mean	2.423	2.423	2.957	2.957	2.617	2.617	3.952	3.952	3.623	3.623	4.667	4.667
CD F statistic		1887.087		641.418		3069.506		1044.129		2990.513		289.605
KP F statistic		11.061		4.435		10.375		5.522		9.342		4.814
Within R^2	0.216		0.266		0.212		0.279		0.248		0.280	
N	28,636	28,636	8,748	8,748	49,110	49,110	13,249	13,249	47,205	47,205	4,033	4,033

Notes: This table re-estimates the benchmark OLS and 2SLS regressions using city population as analytic weights. The weight is constant across drug products within each city-year. All other specifications remain unchanged. Standard errors are clustered at the city level and reported in parentheses. Dep. Var. Mean reports the unweighted mean of the dependent variable in the estimation sample, and within R^2 is reported for OLS specifications only. CD and KP F statistics report the corresponding first-stage instrument-strength tests. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

G. Temperature Inversion as the Sole Instrument

Table OA7: Per-Capita Drug Expenditure: Respiratory Diseases

	Modern			Traditional		
	(1) OLS	(2) First Stage	(3) 2SLS	(4) OLS	(5) First Stage	(6) 2SLS
ln(PM2.5)	0.952*** (0.251)		1.554* (0.788)	1.623*** (0.267)		3.789** (1.551)
Temperature Inversion		0.599*** (0.196)			0.694** (0.237)	
ln(Precipitation)	0.196** (0.078)	-0.091** (0.042)	0.396** (0.162)	0.415*** (0.128)	-0.139* (0.070)	1.065** (0.401)
ln(Humidity)	-0.181 (0.233)	0.165 (0.122)	0.410 (0.456)	-0.274 (0.393)	0.272 (0.157)	-0.342 (0.764)
ln(Temperature)	0.516*** (0.200)	-0.650*** (0.125)	1.447* (0.720)	0.894*** (0.225)	-0.554*** (0.156)	2.889** (1.230)
ln(GDP per capita)	2.638*** (0.326)	0.118*** (0.029)	5.224*** (0.415)	2.713*** (0.416)	0.096*** (0.028)	5.164*** (0.693)
ln(Fiscal revenue)	0.282 (1.763)	-0.973 (0.862)	0.045 (3.297)	0.830 (2.259)	-1.069 (0.991)	0.691 (4.577)
ln(Insurance)	0.216*** (0.059)	-0.081*** (0.023)	0.512*** (0.131)	0.320*** (0.072)	-0.098*** (0.025)	0.801*** (0.212)
ln(Hospital)	1.161** (0.477)	0.123 (0.180)	1.741* (0.995)	0.121 (0.487)	0.151 (0.240)	-0.407 (0.763)
Constant	-12.243** (4.793)	4.913*** (1.201)		-8.514** (3.613)	4.031* (1.782)	
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Var. Mean	2.423	3.914	2.423	2.957	3.914	2.957
CD F statistic			2169.610			1008.499
KP F statistic			9.531			8.505
R ²	0.224	0.950	0.217	0.268	0.944	0.253
N	28,636	29,220	28,636	8,748	8,926	8,748

Notes: This table re-estimates the benchmark regressions using temperature inversion as the sole excluded instrument for ln(PM_{2.5}). All other specifications remain unchanged. Standard errors are clustered at the city level and reported in parentheses. CD and KP F statistics report the corresponding first-stage instrument-strength tests. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table OA8: Per-Capita Drug Expenditure: Cardiovascular Diseases

	Modern			Traditional		
	(1) OLS	(2) First Stage	(3) 2SLS	(4) OLS	(5) First Stage	(6) 2SLS
ln(PM2.5)	0.829*** (0.272)		1.835* (1.052)	1.998*** (0.309)		3.168* (1.627)
Temperature Inversion		0.563*** (0.190)			0.784** (0.242)	
ln(Precipitation)	0.204*** (0.078)	-0.096** (0.041)	0.425** (0.184)	0.416** (0.186)	-0.135* (0.069)	0.945 (0.512)
ln(Humidity)	-0.145 (0.252)	0.186 (0.124)	0.148 (0.405)	-0.798* (0.451)	0.308 (0.200)	-0.326 (0.874)
ln(Temperature)	0.573*** (0.202)	-0.652*** (0.122)	1.547* (0.863)	1.284*** (0.239)	-0.517*** (0.125)	2.428* (1.172)
ln(GDP per capita)	2.696*** (0.342)	0.122*** (0.027)	5.568*** (0.490)	3.260*** (0.610)	0.093** (0.036)	6.541*** (0.873)
ln(Fiscal revenue)	-0.316 (1.857)	-1.064 (0.848)	-0.119 (3.479)	-1.886 (2.522)	-0.919 (0.824)	-1.987 (5.260)
ln(Insurance)	0.248*** (0.066)	-0.077*** (0.023)	0.550*** (0.148)	0.490*** (0.058)	-0.089*** (0.017)	0.923*** (0.201)
ln(Hospital)	1.138** (0.510)	0.112 (0.179)	1.768* (0.983)	0.331 (0.463)	0.049 (0.224)	0.711 (0.928)
Constant	-11.624** (4.866)	4.922*** (1.233)		-10.196*** (3.677)	4.367** (1.875)	
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Var. Mean	2.617	3.925	2.617	3.952	3.901	3.952
CD F statistic			3286.527			1804.021
KP F statistic			8.962			10.701
R ²	0.198	0.949	0.214	0.267	0.942	0.270
N	49,110	50,112	49,110	13,249	13,519	13,249

Notes: This table re-estimates the benchmark regressions using temperature inversion as the sole excluded instrument for ln(PM_{2.5}). All other specifications remain unchanged. Standard errors are clustered at the city level and reported in parentheses. CD and KP F statistics report the corresponding first-stage instrument-strength tests. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table OA9: Per-Capita Drug Expenditure: Tumors

	Modern			Traditional		
	(1) OLS	(2) First Stage	(3) 2SLS	(4) OLS	(5) First Stage	(6) 2SLS
ln(PM2.5)	1.268*** (0.375)		2.601* (1.343)	1.928*** (0.364)		3.926* (1.805)
Temperature Inversion		0.527** (0.194)			0.762** (0.248)	
ln(Precipitation)	0.274** (0.119)	-0.080* (0.042)	0.530** (0.230)	0.567** (0.227)	-0.102 (0.072)	1.004** (0.431)
ln(Humidity)	-0.217 (0.327)	0.145 (0.124)	0.206 (0.500)	-0.712* (0.365)	0.221 (0.207)	-1.140 (0.888)
ln(Temperature)	0.863*** (0.272)	-0.684*** (0.116)	2.192* (1.118)	1.072*** (0.326)	-0.550*** (0.118)	2.889* (1.350)
ln(GDP per capita)	3.506*** (0.504)	0.131*** (0.032)	6.381*** (0.649)	3.707*** (0.742)	0.086* (0.042)	7.020*** (0.916)
ln(Fiscal revenue)	1.473 (2.195)	-0.970 (0.794)	1.620 (4.218)	2.274 (3.393)	-0.910 (0.854)	0.785 (6.029)
ln(Insurance)	0.379*** (0.067)	-0.070*** (0.022)	0.696*** (0.164)	0.669*** (0.068)	-0.093*** (0.018)	1.146*** (0.212)
ln(Hospital)	1.572*** (0.557)	0.117 (0.165)	2.407* (1.150)	1.392* (0.760)	0.062 (0.252)	1.252 (1.117)
Constant	-16.849*** (5.337)	5.132*** (1.087)		-16.769*** (5.290)	4.707* (2.162)	
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Var. Mean	3.623	3.923	3.623	4.667	3.886	4.667
CD F statistic			2717.948			545.998
KP F statistic			7.551			11.827
R ²	0.247	0.951	0.256	0.265	0.945	0.271
N	47,205	48,167	47,205	4,033	4,115	4,033

Notes: This table re-estimates the benchmark regressions using temperature inversion as the sole excluded instrument for ln(PM_{2.5}). All other specifications remain unchanged. Standard errors are clustered at the city level and reported in parentheses. CD and KP F statistics report the corresponding first-stage instrument-strength tests. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

H. Migration and Pollution

Table OA10: Air Pollution and Migration in 2015

	(1)	(2)	(3)	(4)	(5)
	Out-migrants	In-migrants	Out-migration rate	In-migration rate	In-migrants
ln(PM2.5), 2015	-0.054 (0.513)	-1.229** (0.549)	-0.000 (0.008)	-0.043** (0.018)	
Low pollution in 2015, defined as -ln(PM2.5)					1.229** (0.549)
Constant	12.245*** (2.002)	16.939*** (2.311)	0.039 (0.031)	0.221*** (0.073)	16.939*** (2.311)
Migration data level	Province/municipality				
Outcome scale	Log count	Log count	Rate	Rate	Log count
R ²	0.001	0.173	0.000	0.110	0.173
N	20	20	20	20	20

Notes: This table examines whether migration-related population selection is systematically associated with air pollution. Using the 2015 migration matrix, columns (1) and (2) regress the log numbers of out-migrants and in-migrants on PM2.5, respectively, while columns (3) and (4) use the corresponding migration rates. Column (5) regresses the log number of in-migrants on low pollution, defined as $-\ln(\text{PM2.5})$. Because migration data are available at the province or municipality level, each sample city is matched to its corresponding province or municipality. The results provide little evidence that more polluted regions experience systematically greater out-migration, while lower-pollution regions tend to receive more in-migrants. Standard errors are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

I. Comparison of Sample Cities with Other Chinese Cities

Table OA11: Characteristics of Sample Cities and Other Chinese Cities

	Statistics				No. of observations	
	20 cities	Other cities	Difference	Ratio	20 cities	other cities
<i>Economic and population</i>						
Fiscal revenue per capita, yuan	17256.42	4579.06	12677.37***	3.77	120	1,639
GDP per capita, 10,000 yuan	10.08	5.22	4.86***	1.93	100	1,361
Population, million	9.94	4.03	5.90***	2.46	120	1,646
<i>Health-care access, city</i>						
Hospital beds per 10,000 residents, city	75.86	46.53	29.33***	1.63	120	1,637
Physicians per 10,000 residents, city	42.89	23.47	19.42***	1.83	120	1,639
Employee medical insurance coverage, pct., city	66.66	23.00	43.66***	2.90	117	1,613
Hospitals per million residents, city	39.55	45.43	-5.88**	0.87	120	1,639
<i>Health-care access, districts</i>						
Hospital beds per 10,000 residents, districts	94.11	76.24	17.86***	1.23	115	1,600
Physicians per 10,000 residents, districts	51.77	36.29	15.48***	1.43	116	1,607
Employee medical insurance coverage, pct., districts	71.47	33.88	37.59***	2.11	111	1,562
Hospitals per million residents, districts	42.29	52.89	-10.60***	0.80	116	1,603

Notes: This table compares characteristics of the 20 sample cities with those of other Chinese cities using city-year observations from 2014–2019. The ratio column reports the mean for the sample cities relative to that for other cities. Statistical significance of differences in means is based on robust standard errors. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Source: China City Statistical Yearbook and authors' calculations.

Table OA12: Characteristics of Sample Cities and Other Cities in the Same Provinces

	Statistics				No. of observations	
	20 cities	Other cities	Difference	Ratio	20 cities	other cities
<i>Economic and population</i>						
Fiscal revenue per capita, yuan	17256.42	4524.72	12731.71***	3.81	120	1,085
GDP per capita, 10,000 yuan	10.08	5.42	4.66***	1.86	100	903
Population, million	9.94	4.45	5.49***	2.23	120	1,085
<i>Health-care access, city</i>						
Hospital beds per 10,000 residents, city	75.86	46.83	29.03***	1.62	120	1,082
Physicians per 10,000 residents, city	42.89	23.58	19.31***	1.82	120	1,084
Employee medical insurance coverage, pct., city	66.66	25.79	40.87***	2.58	117	1,065
Hospitals per million residents, city	39.55	39.66	-0.11	1.00	120	1,084
<i>Health-care access, districts</i>						
Hospital beds per 10,000 residents, districts	94.11	78.31	15.80***	1.20	115	1,067
Physicians per 10,000 residents, districts	51.77	37.00	14.77***	1.40	116	1,069
Employee medical insurance coverage, pct., districts	71.47	38.19	33.29***	1.87	111	1,035
Hospitals per million residents, districts	42.29	51.50	-9.21***	0.82	116	1,071

Notes: This table compares characteristics of the 20 sample cities with those of other cities in the same provinces or municipalities using city-year observations from 2014–2019. The ratio column reports the mean for the sample cities relative to that for other cities in the same provinces or municipalities. Statistical significance of differences in means is based on robust standard errors. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Source: China City Statistical Yearbook and authors' calculations.